

Stock Price Prediction using LSTM Neural Network: A Case Study Using Safaricom Stocks

Victor Radeny

iLab Africa, Strathmore University, Nairobi, Kenya

Introduction

Background: Stock price prediction is a critical challenge for investors. Traditional methods like technical and fundamental analysis often fall short due to their inability to handle complex, non-linear patterns in stock prices. This study proposes a multivariate LSTM model to predict stock prices using technical indicators and lagged prices as inputs to the model.

Methodology

Data: The dataset consists of Safaricom PLC (SCOM) stock data from 2010 to 2024, including Open, Close, High, Low, and Volume.

Model: A multivariate LSTM model was developed using 20 features selected via Recursive Feature Elimination (RFE) down from 44 initial features. We then ran two model types:

- **Arbitrary LSTM:** A basic LSTM model with default hyperparameters.
- **Tuned LSTM:** A fine-tuned LSTM model with optimized hyperparameters.

Evaluation: Model performance was assessed using MSE, RMSE, MAE, MAPE, and R-squared metrics.

Results

- The tuned LSTM model outperformed the arbitrary LSTM model across all metrics.
- The tuned model achieved an R-squared of 0.7562 and a MAPE of 0.1199.
- The model was able to follow stock price trends but struggled with sudden price shocks.

Model Performance

Metric	Tuned LSTM	Arbitrary LSTM
MSE	9.3109	21.2153
RMSE	3.0514	4.6060
MAE	2.2803	3.4543
MAPE	0.1199	0.1763
R-squared	0.7562	0.4445

Table: Comparison of Tuned and Arbitrary LSTM Performance

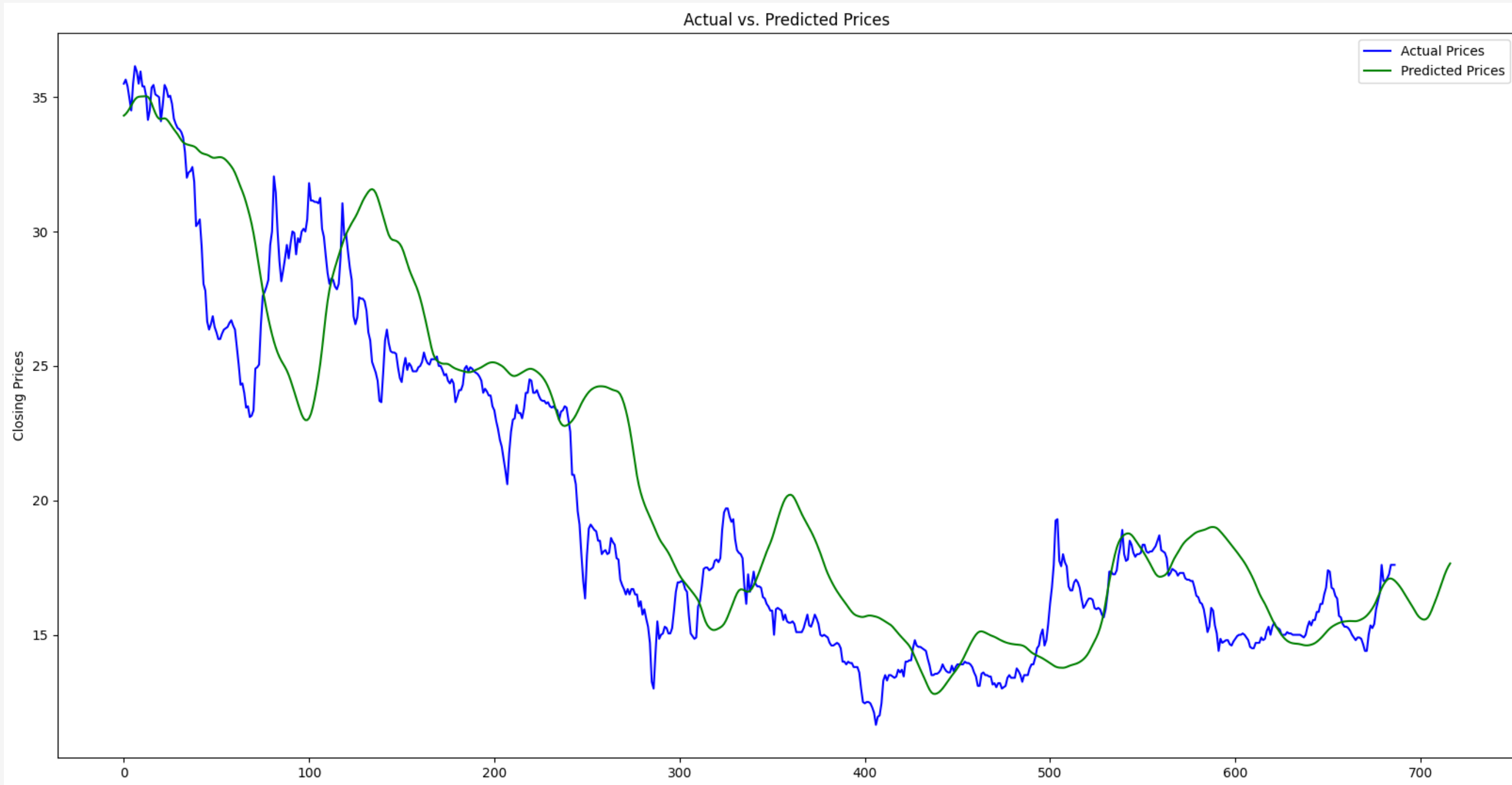


Figure: LSTM-Predicted Prices

Conclusion

Findings: The study demonstrates that a well-constructed LSTM model can significantly enhance short-term stock price predictions. The tuned model outperformed the arbitrary model, highlighting the importance of hyperparameter tuning.

Future Work: Future research could explore the integration of external factors like market sentiment to improve prediction accuracy.

Contact: victor.radeny@strathmore.edu — **Source Code:** GitHub Repository